

Grover's Search Algorithm and Amplitude Amplification

Session 5 – Day 2

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- Compare the query complexities of classical vs. quantum unstructured search.
- Generalize Grover's search to arbitrary state preparation via Amplitude Amplification.

The Search Problem and Quantum Speedup

- **Problem:** Search an unstructured database of size $N = 2^n$ to locate a unique target state $|\omega\rangle$ that satisfies a black-box function $f(x)$:

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- Classical expected queries: 500,000.
- Grover's quantum search queries: $\approx \frac{\pi}{4} \sqrt{10^6} \approx 785$.
- This represents a roughly 600 $\times$ query reduction.

The Oracle and Phase Kickback

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$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle$$

- Since the target register remains in $|-\rangle$, we can view U_f as an n -qubit phase oracle:

$$U_\omega |x\rangle = (-1)^{f(x)} |x\rangle = \begin{cases} -|x\rangle & \text{if } x = \omega, \\ |x\rangle & \text{if } x \neq \omega. \end{cases}$$

$$U_\omega = I - 2|\omega\rangle\langle\omega|$$

The Grover Iteration

We start with the uniform superposition state:

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- 1 **Phase Oracle (U_ω):** Flips the sign of the target state:

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- 2 **Grover Diffusion Operator (D):** Reflects the state vector about the uniform superposition state $|s\rangle$:

$$D = 2 |s\rangle \langle s| - I = H^{\otimes n} (2 |0^n\rangle \langle 0^n| - I) H^{\otimes n}$$

This operator performs an **inversion about the mean**, mapping each amplitude α_x to $2\mu - \alpha_x$, where μ is the average amplitude.

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All 8 amplitudes are $\alpha_x = 0.354$. Initial success probability is $1/8 = 12.5\%$.

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- ❷ **Subspaces:** Define the good and bad states:

$$|\psi_{\text{good}}\rangle = |011\rangle, \quad |\psi_{\text{bad}}\rangle = \frac{1}{\sqrt{7}} \sum_{x \neq 011} |x\rangle$$

$$|\psi\rangle = \sin \theta |\psi_{\text{good}}\rangle + \cos \theta |\psi_{\text{bad}}\rangle$$

where $\sin \theta = \frac{1}{\sqrt{8}} \approx 0.354 \implies \theta \approx 20.7^\circ$.

Toy Example: First Grover Iteration

- ① **Apply Oracle U_f :** Flips the sign of the target element $|011\rangle$:

$$\alpha_{011} \rightarrow -0.354, \quad \alpha_x \rightarrow +0.354 \quad (x \neq 011)$$

The new mean amplitude is:

$$\mu' = \frac{7 \times (0.354) - 0.354}{8} = \frac{6 \times 0.354}{8} \approx 0.265$$

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- ③ **Success Probability:** $P(011) = (0.884)^2 \approx 78.1\%$ (up from 12.5%).

Toy Example: Second Grover Iteration

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- ③ **Success Probability:** $P(011) = (0.970)^2 \approx 94.1\%$.

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Rotation Angle for N

$$\theta = \arcsin(1/\sqrt{8}) \approx 20.7^\circ \implies 2\theta \approx 41.4^\circ$$

Starting at angle $\theta \approx 20.7^\circ$, after 2 iterations we are at:

$$20.7^\circ + 2 \times (41.4^\circ) = 103.5^\circ$$

Grover's Search: Complexity and Overshooting

- After k iterations, the state of the system is:

$$G^k |\psi\rangle = \sin((2k+1)\theta) |\psi_{\text{good}}\rangle + \cos((2k+1)\theta) |\psi_{\text{bad}}\rangle$$

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Oscillation of Success Probability for $N = 8$:

Iterations (k)	Angle $((2k+1)\theta)$	Success Probability
0	20.7°	12.5% (initial)
1	62.1°	78.1%
2	103.5°	94.4% (optimal)
3	144.9°	33.2% (overshot)

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with a success probability of $p_{\text{good}} = \sin^2\theta$.

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- Applying Q rotating by 2θ each time allows us to find a good solution in $\mathcal{O}(1/\sqrt{p_{\text{good}}})$ queries.
- This is a quadratic speedup over classical search, which requires $\mathcal{O}(1/p_{\text{good}})$ trials.

Amplitude Amplification: Toy Examples

Example 1: Success Probability $p = 1/4$

$$\sin \theta = \sqrt{1/4} = 1/2 \implies \theta = 30^\circ.$$

$$\text{Optimal iterations: } k \approx \frac{\pi}{4\theta} = 1.5 \implies k = 1.$$

After 1 iteration, success probability is $\sin^2(3\theta) = \sin^2(90^\circ) = 1.0$.

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Example 2: Success Probability $p = 1/2$

$$\sin \theta = \sqrt{1/2} \implies \theta = 45^\circ.$$

After 1 iteration ($k = 1$), success probability is $\sin^2(3\theta) = \sin^2(135^\circ) = 1/2$.

A single iteration does not improve the probability because the state rotates past $|\psi_{\text{good}}\rangle$.

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Discussion Problem 1: Unitarity of the Diffuser

Problem: Prove that the Grover diffuser $D = 2|s\rangle\langle s| - I$ is unitary ($D^\dagger D = I$) and that it acts as a reflection. **Proof:**

- 1 Since the state $|s\rangle$ is normalized, we have $\langle s|s\rangle = 1$.
- 2 D is self-adjoint:

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- 4 For any state $|\psi\rangle = a|s\rangle + b|s_\perp\rangle$:

$$D|\psi\rangle = a(2|s\rangle\langle s|s\rangle - |s\rangle) + b(2|s\rangle\langle s|s_\perp\rangle - |s_\perp\rangle) = a|s\rangle - b|s_\perp\rangle$$

This flips the component perpendicular to $|s\rangle$, reflecting the state about $|s\rangle$.

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Problem: For $N = 8$ and target $w = 011$, compute θ explicitly and verify numerically that the amplitude of $|w\rangle$ after $k = 2$ Grover iterations matches $\sin(5\theta)$.

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- This matches our step-by-step amplitude analysis where the target amplitude $\alpha_{011} \approx 0.970$ was obtained.

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④ **Success probability after 3 iterations:**

$$P_{\text{success}} = \sin^2(7\theta) = \sin^2(7 \times 14.48^\circ) = \sin^2(101.36^\circ) \approx 96.1\%$$

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- 3 The algorithm can overshoot the target state if run too long, meaning the number of iterations must be chosen carefully.
- 4 Amplitude amplification generalizes Grover's algorithm to any state preparation process, turning a classical trial success rate of p into a quantum scaling of $\mathcal{O}(1/\sqrt{p})$.